

Module 01

Introduction to AI-Driven Software Quality Assurance (SQA)

Q1. What is Software Quality Assurance (SQA)

Ans: SQA is a set of proactive process maintaining the SDLC to ensure the product meets defined quality and prevent defects before they occur. It focuses on the software reliability, security, customer satisfaction.

Quality is important in software development because it:

- Increase brand reputation and trust
- Reduce development cost
- Increase safety, security and reliability
- Reduce software defects
- Improve customer satisfaction

Q2. Difference between QA vs QC

Quality assurance (QA)	Quality control (QC)
QA focus on preventing defects	QC focuses on finding defects
It improves the software development process	It checks the quality of final product
It is process oriented activity	It is product oriented activity
It happens in SDLC	It happens during testing phase
Ex: establishing coding standards, reviewing requirements	Ex: Running test case to find bugs

Difference between QC vs Testing

Quality control (QC)	Testing
QC is an activity that ensure product quality	Testing means test the software performance
It is product oriented	It is execution oriented
It check whether product meets requirements	It checks whether software behaves correctly
It focus on the finished product by review, inspection, testing	It focus on executing code to find bugs
Ex: code review, documentation review	Ex: functional testing, regression testing

Difference between QA vs Testing

Quality assurance (QA)	Testing
It prevent defects	It finds defects
It covers full SDLC process	It happens after development stage
It focus on process improvement	It focus on product validation
It includes standards, audits, process	It includes manual and automated testing
Ex: creating a secure development process	Ex: checking if the login system works correctly

Q3. How is AI changing Software Testing?

Ans: AI is transforming SQA from manual to automated and faster.

The core areas of transformation are:

- **Automated Test Generation:** Using Natural Language Processing (NLP) AI can read user stories, code diffs or requirements and automatically generate test cases and automation scripts.
- **Self-Healing Test Automation:** In Traditional way if a developer changed a UI button's ID, automated tests would break. But AI tools can now dynamically adapt to minor UI changes and heal the test script automatically.
- **Predictive Defect Analytics:** AI can analyze historical data to predict where bugs are most likely to occur in upcoming releases, allowing teams to focus testing efforts on high-risk areas.
- **Visual Regression Testing:** AI computer vision can scan user interfaces like a human eye, detecting visual bugs that traditional code-based testing misses.
- **Faster Root Cause Analysis:** When a test fails, AI can parse through logs and immediately suggest the exact line of code or environment issue that caused the failure.

Q4. Responsibilities and Daily Activities of QA Engineer

- Responsibilities: Designing test plans, ensuring quality standards across the SDLC, reviewing requirements for testability, and ensuring processes are followed.
- Daily Activities: Participating in sprint planning, reviewing documentation, conducting process audits, tracking quality metrics, and collaborating with developers on best practices.

Responsibilities and Daily Activities of Software Test Engineer

- Responsibilities: Executing test plans, writing test scripts, identifying defects, and verifying fixes. A Software Development Engineer focuses heavily on coding automation frameworks.
- Daily Activities: Writing and maintaining automated test scripts using Selenium, Cypress, Playwright, running regression suites, logging detailed bug reports in tools like Jira, and performing exploratory manual testing.

Responsibilities and Daily Activities of AI Quality Engineer

- Responsibilities: A specialized, emerging role focused on testing AI models like LLMs or machine learning algorithms and leveraging AI tools to test traditional software.
- Daily Activities: Evaluating AI output for accuracy and bias, writing complex prompts to test edge cases, tuning AI-driven test automation tools, and testing integrations between core software and AI APIs.

Q5. Career Roadmap of a QA Professional

